



Mark Scheme (Results)

January 2026

Pearson BTEC National Level 3

Applied Science/Forensic & Criminal Investigation

Unit 1: Principles and Applications of Science I –
Physics (31617H/1P)

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Question Number	Answer	Additional Guidance	Mark
1(a)(i)	A 2 cm (1)		(1)
1(a)(ii)	40 (cm) (1)	Accept forty	(1)
1(b)	C 180° (1)		(1)
1(c)	• Conversion (1) $4.0 \text{ ms} = 4.0 \times 10^{-3} \text{ (s)}$ • Substitution (1) $f = \frac{1}{4.0 \times 10^{-3}}$ • Evaluation (1) 250 (Hz)	250 (Hz) with no working gains full marks Accept 0.004 Allow ECF	(3)
Question Total = 6 marks			

Question Number	Answer	Additional Guidance	Mark
2(a)	A first (1)		(1)
2(b)(i)	- Frequency (of the sound wave) increases (1) (Increasing the tension) increases the wave speed (1) - Frequency is proportional to wave speed (1)	Ignore references to faster vibrations Accept $v = f\lambda$ (and λ is constant) Reject references to wavelength changing	(3)
2(b)(ii)	- Substitution (1) $158 = \sqrt{\frac{T}{2.35 \times 10^{-3}}}$ - Square both sides (1) $24964 = \frac{T}{2.35 \times 10^{-3}}$ - Rearrangement (1) $T = 24964 \times (2.35 \times 10^{-3})$ - Evaluation (1) $T = 58.7 \text{ (N)}$	Correct answer with no working gains full marks Accept ECF Accept correctly rounded answer	(4)

2(c)	• The (incident and reflected) waves travel in opposite directions (along the guitar string) (1) • Forming nodes and antinodes (1)	Accept nodes and antinodes correctly labelled on Figure 4 Accept producing regions of destructive and constructive {superposition/interference}	(2)
Question Total = 10 marks			

Question Number	Answer	Additional Guidance	Mark
3(a)(i)	Gamma (rays) (1)	Accept γ	(1)
3(a)(ii)	Any one linked pair from: • Frequency hopping (1) • Range of frequencies/bandwidth used (1) OR • Pairing when connecting devices (1) • (Pairing) equips devices with security/ identification keys (1)	Accept channel hopping Accept channels Accept rapidly/ constantly changing (frequency) Accept handshaking If no other mark scored, accept Bluetooth has a short distance range (and therefore less chance of devices interfering) (1)	(2)

3(b)	• Cladding around the core has a lower refractive index than the core (1) • Cladding has a lower optical density (1) • Angle of incidence is greater than the critical angle (1)	ORA MP3 is dependent on either MP1 or MP2 being awarded	(3)
Question Total = 6 marks			

Question Number	Answer	Additional Guidance	Mark
4(a)	• (Nearest) base station receives signal from one mobile phone (1) • Transmits the signal (from the base station) to {the other mobile phone/to another base station} (and then to the other mobile phone) (1)	Accept cell tower/ mast/antenna Ignore references to satellites	(2)

Question Number	Indicative Content	Mark
4(b)	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content, but learners should be rewarded for other relevant answers.</i> Both types of waves • Radio waves and microwaves can both be received and transmitted by the low orbit satellite • Longer wave lengths are refracted more in the ionosphere • Radio waves have longer wavelengths than microwaves • Radio waves have lower frequencies than microwaves. Radio waves • Radio waves are reflected/refracted by the ionosphere • Radio waves do not need to use satellite for communication • Radio waves are not affected by wet weather • Radio waves can be received beyond the horizon. Microwaves • Microwaves can travel through the ionosphere • Only microwaves can be received and transmitted by the high orbit satellite • Microwaves are strongly absorbed by water/affected by wet weather • Satellite communication can be disrupted by wet weather.	(6)

Level	Mark	Descriptor – Discuss All traits carry equal weighting
0	0	No rewardable content.
1	1-2	- Adequate interpretation, analysis and/or evaluation of the scientific information with generalised comments being made. - Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. - The discussion shows some structure and coherence.
2	3-4	- Good analysis, interpretation and/or evaluation of the scientific information. - Lines of argument mostly supported through the application of relevant evidence. - The discussion shows a structure which is mostly clear, coherent and logical.
3	5-6	- Comprehensive analysis, interpretation and/or evaluation of all pieces of scientific information. - Line(s) of argument consistently supported throughout by sustained application of relevant evidence. - The discussion shows a well-developed structure which is clear, coherent and logical.
Question Total = 8 marks		